

# Priyal Patel

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## Professional Summary

Artificial intelligence and machine learning engineer with experience building secure, scalable ML platforms and high-performance backend systems. Proficient in Python, Java, Rust, and TypeScript, with expertise in model pipelines, distributed systems, cloud-native infrastructure, and API design. Proven track record of optimizing workflows, improving system performance at scale, and collaborating across teams to deliver production-ready AI solutions.

## Work Experience

### BlackRock, New York NY

AI/ML Engineer II

November 2025 - Present

- Designed and worked on an end to end implementation of Identity and Access Management for the firm's AI/ML platform in **Rust, Typescript, React** and **Tailwind**
- Developed authentication methods for the team's SDK in **Python**.
- Designed and implemented an AI assistant using **RAG** and **Vector DB** implementation

Software Engineer I - Rotational Program Analyst

August 2023 - November 2025

- Part of BlackRock's 2 year rotational analyst program that allows new graduates to rotate through four different teams.
- Mentored 3 summer interns and participated in a reverse mentorship program and panel to promote inclusivity.
- Won the Global Hackathon at BlackRock based on a project that leveraged LLMs to expedite productivity through task and message management.
- Portfolio Management:** Enhanced APIs using **Java, Protobuf** and **gRPC**, improved **Angular/TypeScript** UI components, and built **Python** automation for Excel data comparisons.
- ETFs:** Designed and optimized **Java/Spring Boot** workflows, implemented caching and multithreading and used **Splunk** to identify bottlenecks, reduced processing times by **75–90%** and improved search accuracy and performance.
- Studio:** Contributed to an [open-source Python project](#), built **React/TypeScript** documentation, developed a full-stack metrics platform (**Go, Grafana, Prometheus, PromQL, React**), and implemented secure **OAuth 2.0** secret management with **HashiCorp Vault**.
- Private Markets NLP:** Built **model evaluation pipelines**, implemented **Kubernetes/Docker** health checks, and rewrote **Elastic search** based entity management **APIs** in **Python**.

### Amazon Fulfillment Technologies and Robotics, New York NY

June 2022 – August 2022

Software Engineer Intern

- Implemented a full-stack AR CoMET application, led stakeholder interviews, feature planning, mockups, and upgraded framework, and delivered a customizable, widget-based dashboard in **React, JavaScript, AWS, Kotlin, and Python**

### BlackRock, New York NY

June 2022 – August 2022

Software Engineer Intern

- Implementation of a full stack application that uses **OpenAI, MongoDB, Jupyter Lab, Go, Kubernetes, TypeScript, gRPC**, and **Docker**.
- The application leveraged **OpenAI apis** and **prompt engineering** to help provide
- Implemented the entire backend of the application to make api calls, process data and store the results in **MongoDB**.

### BraceBridge Capital, Boston MA

December 2021 – May 2022

Software Engineer Co-op

- Refactored and optimized a core **C#** and **T-SQL** library using **OOD** principles and design patterns, improving runtime performance by 10–30% and enhancing stored procedures.
- Built a full-stack Blazor dashboard to support execution, testing, and monitoring, while migrating dependencies from **.NET Core** to **.NET** and reducing reliance on legacy libraries.

## Education

### Northeastern University, Boston, MA

September 2020- May 2023

Master of Science, Computer Science, GPA: 3.97 /4.0

### Monmouth University, West Long Branch, NJ

September 2012 – May 2017

Bachelor of Science, Biology, Molecular and Cell Physiology, GPA: 3.97/4.0

Bachelor of Science, Medical Laboratory Sciences, GPA: 3.95/4.0

## Technical Skills

**Languages:** Java, Python, C, HTML, CSS, JavaScript, C#, Blazor, C#, Kotlin, NodeJS, React, Redux, Express, Go, TypeScript, Rust

**Databases:** T-SQL/MS SQL, Firebase, MongoDB, SQLite (python), MySQL, Postgres, PromQL

**Operating Systems/Game engines:** MacOS, Windows 10, Linux (Ubuntu), Unity

## Certifications

**Azure Fundamentals**, Microsoft, [Credential ID: 90E644094EA7D699](#)

March 2025